

Salt Marsh Vegetation at NCOS

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Guiding questions

- Which dominant plant species characterize salt marsh habitat at NCOS?
- How has the mean percent cover of the dominant salt marsh species changed across survey years?
- How has the relative contribution of native and non-native vegetation changed across survey years?

1. Set up

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.1.5  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.1      v tibble     3.3.0  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
# file organization  
library(here)
```

here() starts at /Users/macey/Desktop/GitHub/ncos-vegetation

```
# data visualization
library(patchwork)
library(naniar)
```

Data

```
veg <- read_csv(here("data", "veg.csv"))
```

Rows: 27036 Columns: 17

-- Column specification -----

Delimiter: ","

chr (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...

dbl (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Rows: 257 Columns: 4

-- Column specification -----

Delimiter: ","

chr (2): year_pool, transect_name

dbl (2): year, num_quad

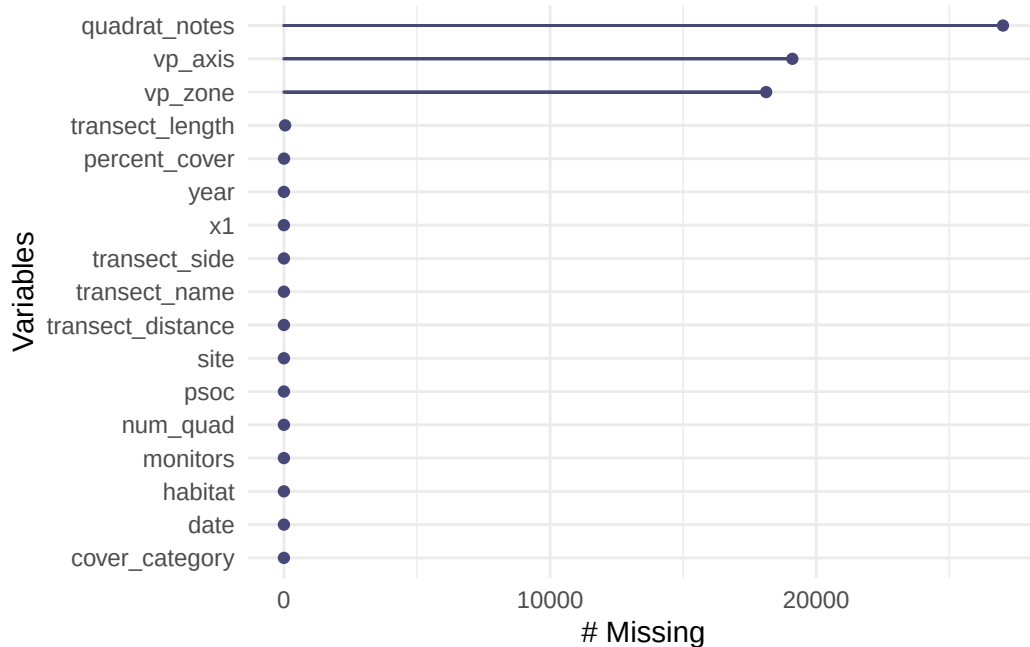
i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

2. Cleaning and wrangling

Explore missingness

```
gg_miss_var(veg)
```



Filter to salt marsh habitat and late summer/early fall surveys

```
saltmarsh <- veg |>
  # filter to salt marsh habitat only
  filter(str_detect(habitat, "Salt Marsh") | str_detect(habitat, "Brackish Marsh")) |>
  # parse date column and filter to August and September
  mutate(date = as.Date(date),
         month = month(date)) |>
  filter(month %in% c(8, 9))
```

Fill implicit NAs with 0

```
saltmarsh_complete <- saltmarsh |>
  complete(year, transect_name, psoc, transect_distance,
         fill = list(percent_cover = 0)) |>
  group_by(year, transect_name, psoc, cover_category) |>
```

```

summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
ungroup() |>
unite("year_pool", year, transect_name, remove = FALSE) |>
left_join(metadata, by = "year_pool") |>
select(-year.y, -transect_name.y) |>
rename(year = year.x,
       transect_name = transect_name.x) |>
mutate(mean_pc = sum_pc / num_quad)

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by year, transect_name, psoc, and cover_category.
- i Output is grouped by year, transect_name, and psoc.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(year, transect\_name, psoc, cover\_category))` for per-operation grouping (`dplyr::dplyr\_by`) instead.

Filter to plant cover categories only

```

saltmarsh_plants <- saltmarsh_complete |>
  filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER"))

```

3. Summarizing for each figure

Figure 3: Species rank (averaged across all years)

```

species_rank <- saltmarsh_plants |>
  group_by(psoc, cover_category) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  arrange(desc(mean_pc))

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by psoc and cover_category.
- i Output is grouped by psoc.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(psoc, cover\_category))` for per-operation grouping (`dplyr::dplyr\_by`) instead.

```
# print to identify dominant species
species_rank
```

```
# A tibble: 104 x 3
```

	psoc <chr>	cover_category <chr>	mean_pc <dbl>
1	Salicornia_pacifica	NATIVE COVER	31.5
2	Distichlis_spicata	NATIVE COVER	17.8
3	Atriplex_lentiformis	NATIVE COVER	12.2
4	Raphanus_sativus	NON-NATIVE COVER	11.6
5	Eleocharis_macrostachya	NATIVE COVER	10.5
6	Festuca_perennis	NON-NATIVE COVER	9.97
7	Distichlis_littoralis	NATIVE COVER	9.96
8	Suaeda_taxifolia	NATIVE COVER	9.88
9	Elymus_triticoides	NATIVE COVER	9.13
10	Isocoma_menziesii	NATIVE COVER	8.4

```
# i 94 more rows
```

Figure 1: Dominant species by year

```
# identify top 5 dominant species from species_rank
dominant_species <- species_rank |>
  slice_max(mean_pc, n = 5) |>
  pull(psoc)

# summarize dominant species cover by year
dominant_by_year <- saltmarsh_plants |>
  mutate(psoc_grouped = if_else(psoc %in% dominant_species, psoc, "Other")) |>
  group_by(year, psoc_grouped, cover_category) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE)) |>
  ungroup()
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by year, psoc_grouped, and cover_category.

i Output is grouped by year and psoc_grouped.

i Use `summarise(.groups = "drop\_last")` to silence this message.

i Use `summarise(.by = c(year, psoc\_grouped, cover\_category))` for per-operation grouping (`dplyr::dplyr\_by`) instead.

Figure 2: Native/non-native proportions by year

```
native_nonnative <- saltmarsh_plants |>
  group_by(year, cover_category) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  group_by(year) |>
  mutate(prop = mean_pc / sum(mean_pc)) |>
  ungroup()
```

`summarise()` has regrouped the output.
i Summaries were computed grouped by year and cover_category.
i Output is grouped by year.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(year, cover\_category))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

Figure 4: Total vegetation cover by year

```
total_cover <- saltmarsh_plants |>
  group_by(year, transect_name) |>
  summarize(total_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  group_by(year) |>
  summarize(mean_total = mean(total_pc, na.rm = TRUE),
            se_total = sd(total_pc, na.rm = TRUE) / sqrt(n())) |>
  ungroup()
```

`summarise()` has regrouped the output.
i Summaries were computed grouped by year and transect_name.
i Output is grouped by year.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(year, transect\_name))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

4. Visualizations

Figure 1: Dominant species percent cover across survey years

```
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc_grouped)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(
    # placeholders - update with actual dominant species names after running
    "Other" = "#B4B2A9"
  )) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"))

fig1
```

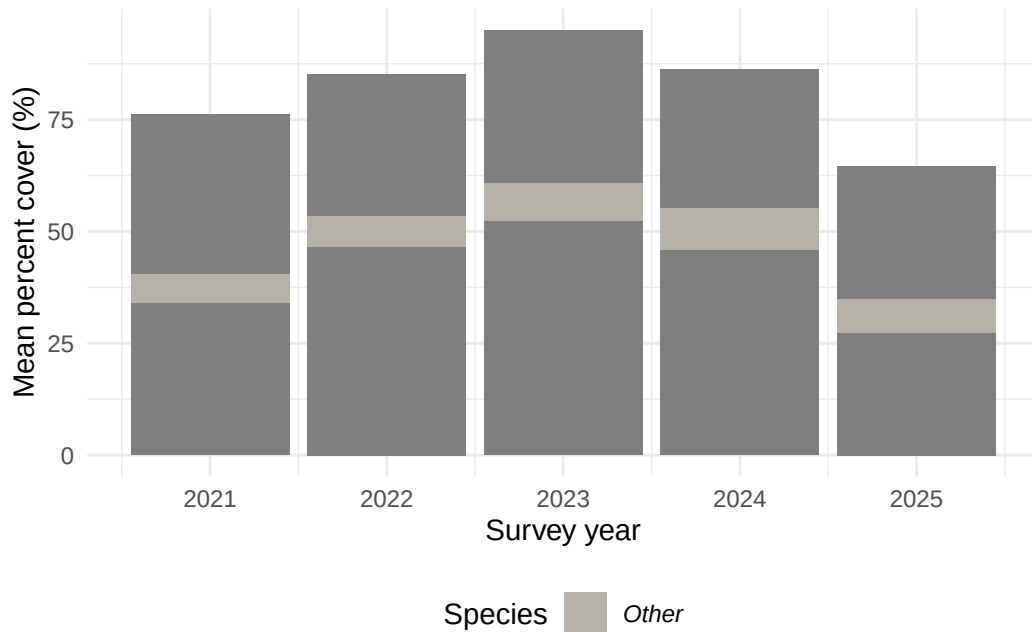


Figure 1: Mean percent cover of dominant salt marsh plant species at NCOS across survey years. The top 5 species by mean cover across all years are shown; remaining species are grouped as Other. Colors indicate native (green) or non-native (coral) status.

Figure 2: Proportional native vs. non-native cover across survey years

```
fig2 <- ggplot(data = native_nonnative,
               aes(x = year,
                   y = prop,
                   fill = cover_category)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Survey year",
       y = "Proportion of vegetation cover",
       fill = "Cover category") +
  theme_minimal() +
  theme(legend.position = "bottom")

fig2
```

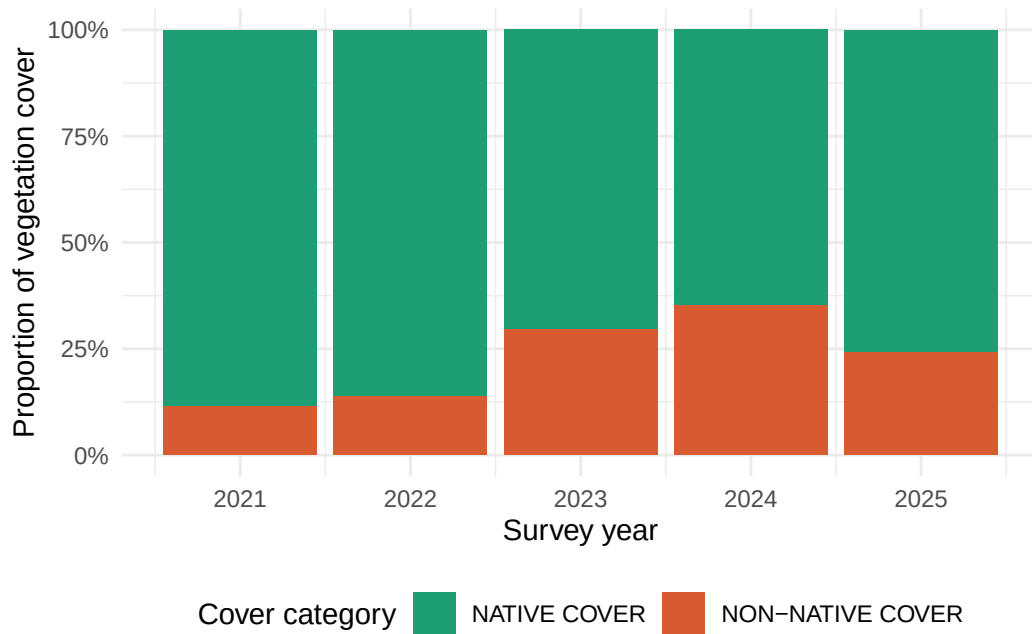



Figure 2: Proportional contribution of native and non-native vegetation cover in salt marsh habitat at NCOS across survey years. Each bar sums to 1. Native cover is shown in green; non-native cover in coral.

Figure 3: Species rank plot

```
fig3 <- ggplot(data = species_rank,
               aes(x = mean_pc,
                   y = reorder(psoc, mean_pc),
                   fill = cover_category)) +
  geom_col() +
  geom_vline(xintercept = species_rank$mean_pc[5],
             linetype = "dashed",
             color = "gray50") +
  scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
  labs(x = "Mean percent cover (%)",
       y = "Species",
       fill = "Cover category") +
  theme_minimal() +
  theme(axis.text.y = element_text(face = "italic"),
```

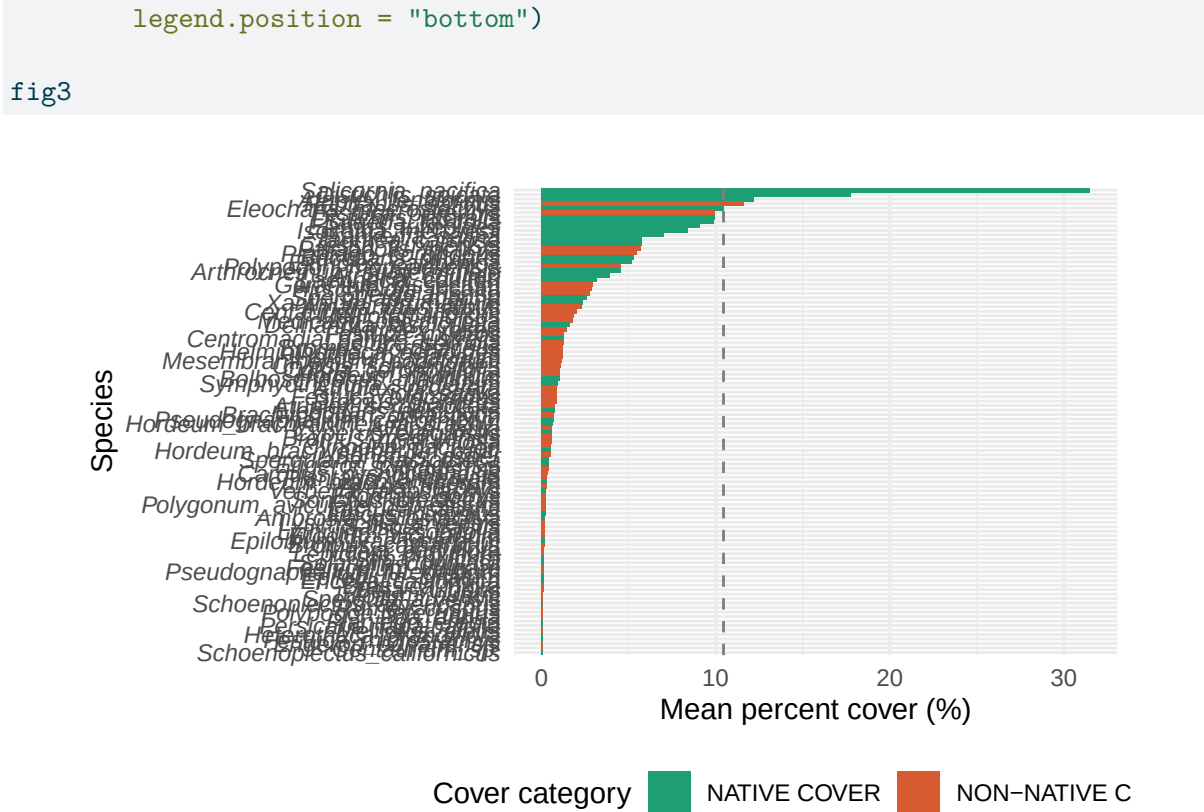


Figure 3: Mean percent cover of plant species recorded in salt marsh habitat at NCOS, averaged across all survey years and ranked from highest to lowest. Green bars indicate native species; coral bars indicate non-native species. Dashed line marks the dominant species cutoff used in figure 1.

Figure 4: Total vegetation cover across survey years

```
fig4 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  geom_ribbon(aes(ymin = mean_total - se_total,
                 ymax = mean_total + se_total),
             fill = "#1D9E75",
             alpha = 0.2) +
  geom_line(color = "#1D9E75",
            linewidth = 1) +
```

```
geom_point(color = "#1D9E75",
           size = 3) +
labs(x = "Survey year",
     y = "Total mean percent cover (%)") +
theme_minimal()
```

fig4

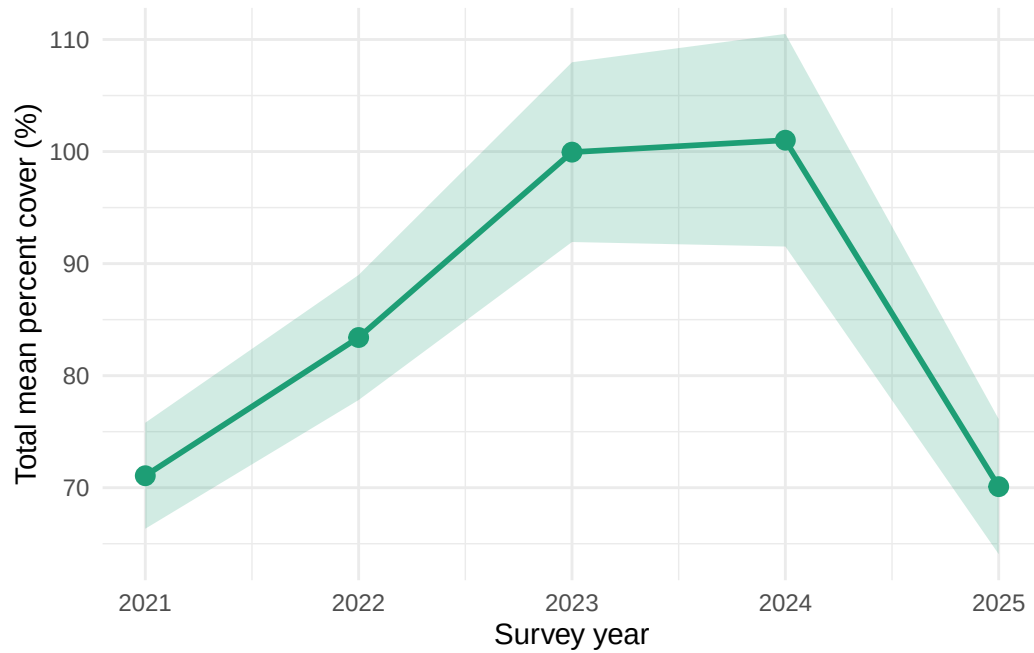


Figure 4: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Shaded band represents ± 1 standard error around the mean.